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Enhancing Consensus for Transdisciplinary Engineering Design Challenges: A Delphi Study Voting Schema

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Abstract. The Delphi method is used to build expert consensus on complex, transdisciplinary issues, yet the selection and reporting of voting schemas remain inconsistent, affecting transparency and comparability. This paper introduces a heuristic framework to guide voting schema selection. A Delphi study with eleven participants compared three vote elicitation and in-study reporting methods, and four consensus analysis methods, to assess the framework's validity. The findings contribute: (1) a novel heuristic, featuring five requirements for voting schema selection in Delphi studies; (2) a recommended schema for socio-technical transdisciplinary challenges, combining a Likert scale, frequency count heat map, and Tastle & Wierman's Consensus and Agreement measures; and (3) practical guidelines for schema selection. This work enhances methodological consistency, transparency and reliability of Delphi studies in transdisciplinary engineering.

Keywords. Consensus-Building, Decision-Making, Participatory.

Introduction

The transdisciplinary nature of modern engineering and design challenges demands robust methods for effective collaboration and decision-making among diverse experts. The Delphi Method, a structured, iterative process that builds consensus through anonymous questionnaires [1], has become a popular tool. Despite the development of tools to enhance Delphi study design over the past years [1-3], little guidance exists on selecting and reporting voting schemas, a critical factor that shapes participant expression and study outcomes [4]. Standardising voting schema design would enable more consistent comparisons across studies and improve the method's utility for addressing complex, real-world challenges. This paper addresses the following research questions:

1. Which voting schema best facilitates expert expression and supports both consensus and dissensus evaluation in Delphi studies investigating emerging, socio-technical, transdisciplinary engineering challenges?
2. How can insights from evaluating different voting schemas inform generalisable guidelines for their selection in transdisciplinary engineering Delphi studies to enhance methodological transparency and reliability?

This work introduces a requirements heuristic for selecting voting schemas (Section 1), applies it to a Delphi study to identify an appropriate schema for emerging, socio-

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technical, transdisciplinary engineering challenges (Section 2), presents the results (Section 3), and offers practical recommendations for future Delphi studies (Section 4).

1. Background: Delphi Method and Voting Schema Requirements

The Delphi Method was developed in the 1950s as a structured process to enable dialogue among experts and build consensus on complex or poorly understood issues while minimising biases arising from social pressures and group influence. A Delphi study is an iterative process where a panel of selected experts independently completes questionnaires, reviews anonymised quantitative and qualitative summaries of previous responses and revises their answers until consensus is reached [1].

The Delphi Method has maintained its core methodological structure while evolving to integrate digital tools, broaden its applications and objectives across diverse fields, adopt a more inclusive definition of expertise, and refine techniques for analysing qualitative and quantitative responses. The method's flexibility, though beneficial in allowing it to be applied to diverse research contexts, has resulted in a lack of standardisation which, combined with inconsistent reporting, makes it challenging to compare findings across studies or establish best practices. To address these challenges several instruments have been proposed in the literature that supports study design decision-making through practical recommendations [1], quality appraisal tools [2], and methodological reviews [3]. While these instruments address many aspects of the Delphi study design, there is a lack of resources to guide voting schema selection and reporting.

A Delphi study's voting schema defines the mechanisms for: ***1) response elicitation***; ***2) analysis***; and ***3) in-study result reporting*** (controlled feedback). It dictates how participants express their opinions, engage with the perspectives of other panel members, and establish criteria for consensus and stability. By structuring the consensus-building and decision-making processes, the voting schema directly shapes the study's outcomes.

While a few comparative studies on voting schema design can be found in the literature [4-7] these are narrow in focus, limited to singular features of a Voting Schema, rather than the voting schema in its entirety. Ehringfield et al. [4] compare the impact of vote elicitation methods using social choice theory, however, provide limited acknowledgement of the role of in-study reporting and consensus analysis methods.

The complex nature of emerging socio-technical transdisciplinary engineering challenges demands a holistic approach to schema design. In such contexts, focusing on isolated aspects of the voting schema risks overlooking the interdependencies that critically shape consensus-building and decision-making.

For a Delphi study to yield valid, credible, and meaningful results, its voting schema must meet the following key criteria:

R1. The schema's elicitation method must enable individuals to meaningfully express nuanced opinions.
R2. The schema's in-study results reporting must validly reflect the panel's views.
R3. The schema's in-study results reporting must be clear, interpretable, and easily understood by all panellists.
R4. The schema's analytical methods must enable the assessment of consensus, dissensus, and in/stability with appropriate levels of stringency, sensitivity and intuitive validity.
R5. The schema's thresholds for determining consensus, dissensus, and in/stability must be clear, justifiable, and methodologically sound.

2. Methodology – A Study of Voting Schema

A multi-round Delphi study was conducted with eleven experts in design, engineering and manufacturing to explore voting schema design for use in emerging socio-technical transdisciplinary engineering challenges. The study employed a case study of the development of a Community Robot for environmental monitoring. This was selected as a case study as it permits the involvement of varied stakeholders and experts, and forms part of a larger project that would seek to apply the findings in this real scenario. The study methodology, described in this section, is summarised in [8].

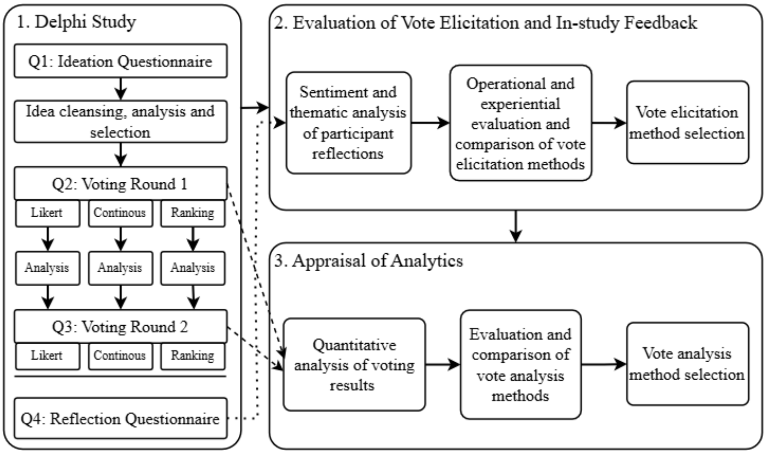

Figure 1. Diagram illustrating the Delphi study and voting schema evaluation methodology. Solid arrows illustrate methodological procedure. Dotted and dashed arrows illustrate the flow of data.

2.1. Study Design

The Delphi study was conducted over several rounds, including an ideation round and two voting rounds, followed by a reflective questionnaire. Eleven participants were recruited to the study panel, all meeting the inclusion criteria of: expertise and experience in engineering, design and/ or manufacturing, and good written and spoken English comprehension. Questions were completed independently, and responses were kept anonymous throughout.

Round 1- Ideation: Participants were introduced to the research scenario, with background information and key definitions, before ideating characteristics defining a waterway monitoring robot as a responsible Community Robot. They were encouraged to consider the entire technology innovation system, encompassing the product itself, the process through which it was developed, and the paradigm of values and outcomes it enables.

The research team aggregated and coded the responses, categorising factors deductively as product-, process-, and/ or paradigm- related while applying inductive thematic coding. Similar factors were consolidated, and statements were refined for clarity, consistency, and conciseness. To provide a representative schema experience while minimising the burden of participation, a subset of 20 statements was selected for voting. These statements, drawn from the generated ideas, span five thematic categories: Beneficiaries, Function, Pollution, Transparency, and Participation.

Round 2 – Voting Round 1: After reviewing the study context, participants rated the importance of the selected statements using three voting methods. They completed each method independently, directed not to cross-reference responses between methods. Participants had the option to provide justifications for their votes. The submitted data was aggregated, and descriptive statistics were calculated for each voting method.

Round 3 – Voting Round 2: For each vote elicitation method, after reviewing the descriptive statistics and justifications from the previous round, participants were invited to re-vote. A summary of the study scenario was provided.

Reflections Questionnaire: Finally, participants reflected on both the voting methods and the presentation of voting history data. Regarding the voting methods, they were asked to indicate their preferred method, identifying which enabled them to best express their opinion, and explaining their reasoning. For the voting history data, they were asked to evaluate and comment on their clarity, and to consider how the displayed vote statistics and/ or aggregated justifications influenced their second-round decisions.

Study data were collected using digital qualitative and quantitative questionnaires. The Delphi study data was collected using digital questionnaires formatted in Microsoft Excel, with custom validation indicators. Participant reflections were elicited and collected via an email exchange.

2.2. Evaluation of Vote Elicitation and In-Study Reporting

Three rating voting methods were identified as potential candidates. The methods selected are the Likert Scale, the Continuous Scale, and Rank-Order Assignment. These methods were selected because they are suitable for collecting rated measures of relative importance (R1), provide methods for the creation of in-study reported descriptive statistics (R2, R3), and have robust mathematical techniques for analysing consensus, dissensus, and stability (R4, R5). Fixed Sum rating, Pick Some ranking, and Paired Comparison methods were discounted due to the challenges associated with using these methods to evaluate and compare voting items across a wide range of topics or themes, and with varying degrees of familiarity.

For each voting method, appropriate vote elicitation methods, method granularity, and reporting formats were selected. These are presented in Table 1 below.

Table 1. Overview of the vote elicitation methods and in-study reporting formats used.

Vote Elicitation Method	Granularity	In-study reporting formats
Likert Scale	7-point scale with options: Not at all important (NI), Low importance (LI), Slightly important (SI), Neutral (N), Moderately important (MI), Very important (VI), Extremely important (EI).	Numerical frequency count and a heat map visualisation, raw justifications.
Continuous Scale	Scale from 0 (Not at all important) to 100 (Extremely important).	Median and interquartile range (0 decimal places), and a box-and-whisker plot, raw justifications.
Rank Order Assignment	Ordered ranking from most important (first) to least important (last) with unique ranks for all characteristics.	Numerical median, minimum, and maximum ranks, raw justifications.

After completing the Delphi study questionnaires, a down-selection process was conducted to evaluate and compare the three schemas based on their operational methodology and experiential feedback (R1, R2, R3). This evaluation was exclusively informed by the results from the Reflections Questionnaire.

2.3. Appraisal of Analytics

Following the selection of the Likert scale as the most suitable vote elicitation method, four analytical methods (M) for measuring consensus and stability were evaluated (R4, R5), each at two threshold levels (T). The methods, which include mathematical assessments of dispersion and central tendency, and defined thresholds for consensus, are detailed in Table 2 and were applied exclusively to the Likert data.

Table 2. Overview of consensus measurement methods (M), including dispersion and central tendency measures with their corresponding thresholds (T).

M	Dispersion Measure	Central Tendency	Consensus Thresholds
M1	Interquartile Range (IQR)	Median	T1. IQR ≤1.5 points T2. IQR ≤1 point
M2	Maximum Percentage Agreement (PA) within 1- and 2-point Likert ranges	Mode	T1,3. $PA_{max} \geq 70\%$ T2,4. $PA_{max} \geq 85\%$
M3	Tastle & Wierman's Consensus Measure, C_m[8] $C_m(X) = \sum_{i=1}^{n} p_i \log_2\left(1 - \frac{\lvert X_i - \mu_X \rvert}{d_X}\right)$	Median	$T1.\ C_m \geq 0.7$ T2. $C_m \geq 0.85$
M4	Maximum Tastle & Wierman's Agreement Measure, A_m [9] $A_m(X,\tau) = \sum_{i=1}^{n} p_i \log_2\left(1 - \frac{\lvert X_i - \tau \rvert}{2d_X}\right)$	Maximum A_m	T1. $A_m \geq 0.7$ T2. $A_m \geq 0.85$

- Where, p_i is the probability of a vote at Likert rating X_i , μ_X is the mean of X, d_X is the width of X, $d_X = X_{max} - X_{min}$, and τ is the target Likert category.
- To calculate the interquartile range, median, C_m, and A_m the Likert scale was mapped to a numerical range (X_i) 1 to 7, with increments of 1.

The selected thresholds align with existing literature, where Delphi study consensus thresholds are set between 70% and 85% and an IQR between 1 and 2 units [2, 10].

The analytical methods were evaluated based on five key criteria: ***i) consensus stringency*** - how liberally the method assigns consensus; ***ii) threshold sensitivity*** - the extent to which variations in the consensus threshold affect the results; ***iii) sensitivity to vote distribution*** - the method's ability to capture the spread and clustering of votes*; iv)* ***alignment with intuitive validity*** - whether the results align with logical and intuitive interpretations of consensus; and, ***v) the resolution of in/stability assessment*** - the capacity to distinguish between stable and unstable responses.

3. Results

3.1. Delphi study outputs

In the first questionnaire, participants generated over 170 ideas spanning the product, process, and paradigm of the robot. Through inductive open coding, these factors were categorised into 32 themes, including beneficiaries, data, end-of-life, function, pollution, and transparency. Panel participation throughout the study is presented in Table 3.

Table 3. Panellist participation in each questionnaire and round of the Delphi study.

Vote Elicitation Method	Q1 Ideation	Q2 Voting 1	Q3 Voting 2	Q4 Reflections
Likert		7	9	
Continuous	11	7	9	10
Rank Order		9	9	

3.2. Evaluation of Vote Elicitation and In-Study Feedback

Following the completion of the Delphi Study, the three voting schemas were evaluated and compared using the qualitative reflections on the Likert Scale, Continuous Scale, and Rank-Order methods gathered from participants in the Delphi study.

Among the three voting methods, five participants expressed a clear preference for the Likert Scale, while three favoured the Rank-Order method. No participants indicated a preference for the Continuous Scale; however, two indicated a preference for a mixed-methods approach.

Participants expressed a preference for the Likert Scale due to its user-friendliness and low cognitive burden, which allowed them to convey their opinions and make comparisons among ratings. Additionally, participants found the Likert Scale familiar and suitable for qualitative assessments of qualitative characteristics.

The Continuous 100-point rating scale was described as cognitively demanding by several participants, with multiple respondents highlighting the difficulty in assigning specific values, calling the process "arbitrary" and "fatiguing."

Participants praised the Rank-Order method for its flexibility in expressing preferences. It was also described as "engaging" and "thorough" encouraging deeper reflection on priorities as participants "physically mov[ed] the factors into an order list." However, participants noted that the method does not account for any magnitude of difference between ranked items and may become fatiguing with higher item numbers.

Participants broadly supported the descriptive statistics, graphical representations, and justifications reporting results of the previous round. Despite varying strategies for using in-study reports amongst participants, comprehension remained high. The frequency count and heat map were particularly favoured for revealing divisions in opinion beyond standard dispersion and central tendency measures.

The 7-point Likert scale is selected as the most suitable vote elicitation method, as it best satisfies R1. Its balanced response granularity allows participants to express their opinions effectively, differentiate between voting items, and make meaningful comparisons across items describing diverse criteria.

3.3. Appraisal of Analytics

An appraisal of the analytical methods for assessing consensus and dissent was conducted using data from the Likert voting rounds. Four consensus evaluation methods, each applied at two threshold levels, were assessed to determine the most suitable combination. The results, presented in Table 4, show panel vote counts alongside consensus analysis using the four methods defined in Table 2 for a selected set of voting items. The table also includes the total number of items (out of 20) that reached consensus (Total Consensus (all items)) under each method and threshold.

Consensus Stringency: Looking at the Total Consensus, for each voting method the number of voting items reaching consensus varied significantly. For example, for the Maximum Agreement Measure (A_m) at a threshold of 0.70, consensus was reached on all 20 voting items in both rounds whereas for maximum Percentage agreement across a 1-point range at a threshold of 70%, consensus was only reached for one item. For the IQR and Consensus Measure (C_m) methods the total number of voting items reaching consensus falls between these in both rounds.

Threshold Sensitivity: Apart from the maximum Percentage agreement method across a 1-point range, all methods exhibit a sensitivity to the change in the threshold

Table 4. Distribution of Likert-scale importance ratings and consensus outcomes for selected voting items across two voting rounds. Frequencies of responses across seven importance levels – Not at all important (NI), Low importance (LI), Slightly important (SI), Neutral (N), Moderately important (MI), Very important (VI) and Extremely important (EI) – are presented. Consensus was assessed using four methods (M1-M4), each incorporating a measure of dispersion and a central tendency indicator. Results from the selected combination method are also reported. Each item's status is evaluated against predefined thresholds (T): items reaching consensus are shown in bold, while those not meeting the threshold are shown in grey. The number of items (out of 20) achieving consensus under each method and threshold is also reported.

Voting Round	Voting Item	Frequency of Rating (Count)							Analysis Results of Consensus Method (Dispersion // Central Tendency) at Threshold										
									M1. IQR // Median		M2. Percentage Agreement // Mode				M3. C_m // Median		M4. Max A_m. // Max A_m		C_m // Max A_m.
		NI	LI	SI	N	MI	VI	EI	T1. ≤1.5 points	T2. ≤1 point	T1. ≥70% in 1-point	T2. ≥85% in 1-point	T3. ≥70% in 2-points	T4. ≥85% in 2-points	T1. ≥0.7	T2. ≥0.85	T1. ≥0.7	T2. ≥0.85	T. ≥0.8
	3	0	0	0	1	1	2	3	**1.5 // VI**	1.5 // VI	43% //EI	43% //EI	**71% // EI**	71% // EI	**0.766 // VI**	0.766 // VI	**0.891 // VI**	**0.891 // VI**	0.766 // VI
	9	2	1	1	1	2	0	0	3 // SI	3 // SI	29% // NI/MI	29% // NI/MI	43% // NI/MI	43% // NI/MI	0.591 // SI	0.591 // SI	**0.814 // SI**	0.814 // SI	0.591 // SI
	11	0	0	2	0	2	2	1	2 // MI	2 // MI	29% // SI/MI/VI	29% // SI/MI/VI	57% // MI/VI	57% // MI/VI	0.674 // MI	0.674 // MI	**0.851 // MI**	**0.851 // MI**	0.674 // MI
1	14	0	0	2	2	1	2	0	2 // N	2 // N	29% // SI/N/VI	29% // SI/N/VI	57% // SI/N	57% // SI/N	**0.712 // N**	0.712 // N	**0.871 // N**	**0.871 // N**	0.712 // N
	16	0	0	0	3	2	1	1	**1.5 // MI**	1.5 // MI	43% // N	43% // N	**71% //N**	71% // N	**0.766 // MI**	0.766 // MI	**0.891 // MI**	**0.891 // MI**	0.766 // MI
	17	0	1	1	0	3	2	0	**1.5 // MI**	1.5 // MI	43% // MI	43% // MI	**71% // MI**	71% // MI	0.664 // MI	0.664 // MI	**0.867 // MI**	**0.867 // MI**	0.664 // MI
		Total Consensus (All Items)							10	3	1	1	7	1	9	1	20	12	1
	3	0	0	0	0	3	4	2	**1 // VI**	**1 // VI**	44% // VI	44% // VI	**78% // VI**	78% // VI	**0.845 // VI**	0.845 // VI	**0.930 // VI**	**0.930 // VI**	**0.845 // VI**
	9	2	1	1	0	5	0	0	3 // MI	3 // MI	56% // MI	56% // MI	56% // MI	56% // MI	0.538 // MI	0.538 // MI	**0.795 // N**	0.795 // N	0.538 // N
	11	0	0	0	3	1	5	0	2 // VI	2 // VI	56% // VI	56% // VI	67% // VI	67% // VI	**0.773 // VI**	0.773 // VI	**0.898 // VI**	**0.898 // VI**	0.773 // VI
2	14	0	2	0	2	3	1	1	**1 // MI**	**1 // MI**	33% // MI	33% // MI	56% // MI	56% // MI	0.624 // MI	0.624 // MI	**0.837 // MI**	0.837 // MI	0.624 // MI
	16	0	0	1	3	5	0	0	**1 // MI**	**1 // MI**	56% // MI	56% // MI	**89% // MI**	**89% // MI**	**0.841 // MI**	0.841 // MI	**0.929 //MI**	**0.929 //MI**	**0.841 // MI**
	17	0	0	0	2	5	1	1	**0 // MI**	**0 // MI**	56% // MI	56% // MI	**78% //MI**	78% //MI	**0.833 // MI**	0.833 // MI	**0.929 // MI**	**0.929 // MI**	**0.833 // MI**
		Total Consensus (All Items)							13	10	1	1	7	4	10	6	20	12	6

definition. In round one the Consensus Measure (C_m) demonstrated the highest sensitivity, with nine items reaching consensus at a threshold of 0.7, reducing to only one at a threshold of 0.85. In round 2, threshold sensitivity reduces for the IQR, 2-point Percentage Agreement, and Consensus Measure (C_m) methods however remains unchanged for maximum Percentage Agreement and Maximum Agreement Measure (A_m) remained unchanged.

Sensitivity to Vote Distribution: The IQR, Consensus Measure (C_m) and maximum Agreement Measure (A_m) all demonstrate some sensitivity to vote distribution. This effect is more sensitive for C_m and A_m. For example, comparing the round 1dispersion measures for voting items 3 and 16, to item 17; while all have an IQR of 1.5, a distinction between the distributions can be seen in C_m and A_m of items 3 and 16 which have identical, mirrored and translated, frequency distributions (C_m = 0.766, A_m = 0.819), and item 17 (C_m = 0.644, A_m = 0.867), whose distribution differs.

Alignment with Intuitive Validity: Analysis by inspection of vote frequency counts indicates strong alignment with an intuitive definition of consensus by the Consensus Measure (C_m) method of dispersion across all voting items. The Agreement Measure (A_m), IQR and percentage agreement methods all present weaker alignment. For example, voting items 11 and 17 in round 1 have A_m of 0.851 and 0.867 respectively indicating high agreement despite wide vote distributions.

The measures of central tendency all demonstrate broad agreement and align with intuitive evaluations from an analysis by inspection. The results indicate maximum Agreement Measure (A_m) may offer a slight improvement over the median or mode, better accounting for spread and skew of distributions. For example, in round 2, voting item 9 was categorised as of Neutral importance by the maximum Agreement Measure, (A_m) method whereas the mode and median categorised it as Moderately important.

Resolution of Stability Assessment: In assessing vote stability across Delphi rounds, the C_m and A_m allow for direct comparison between rounds and voting items with fine granularity. In contrast, the IQR, restricted to multiples of 0.5, provides a less granular interpretation of stability. For example, voting items 9 and 11 show no change in IQR (IQR = 3 and 2, respectively), suggesting stability. However, changes in C_m and A_m indicate instability, with item 9 showing increasing dissensus and item 11 greater consensus. For the Percentage agreement method, variations in total vote count between rounds impacting the effect size of each vote, make direct comparisons invalid.

4. Discussion

This paper outlined five key requirements (R1–R5) for designing Delphi study voting schemas. The requirements heuristic was applied to identify the voting schema that most effectively facilitates expert expression and enables the exploration and evaluation of both consensus and dissent in Delphi studies on emerging, socio-technical, transdisciplinary engineering challenges. This is found to be a Likert scale, with three feedback features of a frequency count heat map, dispersion and central tendency indicators, and aggregate justifications, and analysis comprising Tastle and Weirman's Consensus Measure (C_m) as a measure of dispersion, maximum Agreement measure (A_m) as the measure of central tendency and consensus and stability thresholds of $Cm \geq 0.8$ and $\Delta C_m \leq 0.04$. This voting schema is elaborated with justification in Table 5.

Table 5. Recommended voting schema for Delphi studies on emerging, socio-technical, transdisciplinary engineering challenges.

	Result	Justification
Elicitation	Likert Scale, 7-point	• Enables expression of nuanced perspectives (R1) • Supports diverse expertise levels and disciplinary backgrounds. • Accommodates different levels of familiarity with the voting topics while ensuring enough granularity for meaningful responses.
In-study Reporting	1. Frequency Count Heat Map 2. Dispersion and Central Tendency Indicators 3. Aggregate justification	• The frequency count provides the clearest view of participant responses (R2). • The color-coded heat map is preferred over other visualisations for clarity (R3). • Measures of dispersion and central tendency provide a high-level representative understanding of the panel position. • Simple stats and justifications help participants understand diverse perspectives and refine their responses.
Analysis	1. Dispersion: Consensus Measure (C_m) 2. Central Tendency: Agreement Measure (A_m) 3. Consensus threshold of $C_m \geq 0.8$ and a stability threshold of $\Delta C_m \leq 0.04$	• No single method is found to be universally superior • Consensus Measure (C_m) intuitively reflects consensus with sensitivity to vote distribution (R4) • Agreement Measure (A_m) provides a clear indication of central tendency, even in skewed or dispersed responses • These thresholds are appropriate for exploring panel consensus and dissensus, while minimising cognitive load (R5).

Considering Delphi studies for transdisciplinary engineering more broadly, to meet R1 the selection of voting methods should consider participant familiarity with the study topic and the comparability of voting item. For topics where participants have high familiarity, a continuous rating scale may be appropriate, allowing for more granular responses. In studies with a narrower scope, where voting items can be directly compared, and the number of items is low to medium, rank-order voting may be suitable. However, given that rank-order methods position each item relative to the others, variation in participant familiarity can affect ranking consistency and overall validity.

Regarding R2 and R3, in-study results reporting should prioritise simplicity to accommodate differences in data literacy among participants. Where more complex analyses or reporting formats are necessary, using multiple presentation formats may enhance accessibility and facilitate engagement across diverse expert groups.

Study results support existing recommendations that thresholds should be set *a priori*, as they significantly influence study outcomes. When establishing thresholds, researchers should consider the study's objective (whether to achieve consensus or explore consensus and dissensus) as well as the cognitive load and fatigue associated with repeated voting on unchanged items, particularly in studies with a high number of voting items. These considerations are especially relevant in the context of emerging transdisciplinary challenges, where the complexity and novelty of the issues may increase cognitive strain on participants.

While the findings of this study are not fully generalisable, they provide valuable insights for designing Delphi studies, particularly those addressing emerging, transdisciplinary engineering and design challenges. The proposed voting schema requirements heuristic has been validated within the scope of this study but is not exhaustive. Nevertheless, it establishes a foundational framework for schema design and reporting, contributing to greater methodological consistency in Delphi studies, enhancing their applicability, transparency, and reliability across research domains.

Several limitations should be acknowledged. The study's participant pool was relatively small and drawn from a limited range of disciplines, which may affect the broader applicability of the findings. Future research should aim to refine and expand the heuristic by testing it across a range of Delphi study objectives, disciplinary contexts, and participant demographics to further strengthen its robustness and generalisability.

5. Conclusion

In this paper, a requirements heuristic to enhance the design of voting schemas and the reporting quality of Delphi studies was introduced. The heuristic was applied in a Delphi study to develop a schema for addressing emerging, socio-technical transdisciplinary engineering challenges. This is found to be a Likert scale, with three feedback features of a frequency count heat map, dispersion and central tendency indicators, and aggregate justifications, and analysis comprising Tastle and Weirman's Consensus Measure (C_m) as a measure of dispersion, maximum Agreement measure (A_m) as the measure of central tendency and consensus and stability thresholds of $C_m \geq 0.8$ and $\Delta\ C_m \leq 0.04$ respectively. The findings complement prior work in social choice theory and Delphi study design, offering practical recommendations for standardising voting schema selection and reporting.

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